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Position

- Assistant Professor, Department of Probability and Statistics, Wuhan University, Hubei, China. June 2022 – Present
- Assistant Professor, Research Institute for Science and Engineering, Waseda University, Japan. Jan 2021– May 2022
- Research Fellow, Risk Management Institute, National University of Singapore, Singapore. June 2020 – Dec 2020
- Research Assistant, Department of Mathematics, National University of Singapore, Singapore. Aug 2019 – June 2020

Education

- Ph.D. in Statistics, Department of Statistics and Applied Probability, National University of Singapore. 2015/09– 2020/03.
 - Thesis topic: Complicated time series modeling & forecasting with high-dimensionality and non-stationarity
 - Supervisor: A/P Ying CHEN. GPA: 4.7/5, A
- B.Sc. in Statistics, Department of Statistics and Finance, University of Science and Technology of China (中国科学技术大学). 2011/09—2015/08.
 - GPA: 91/100 (Top 3%), Outstanding Student Scholarship (Gold) (Top 5%), Zhang Zongzhi Sci-tech Scholarship (Top 5%), Samsung Scholarship (Top 3%)

Research Interests

Functional Data Analysis; Network Analysis; Non-Stationary Time Series Analysis and Forecasting; Count Time Series.

Funding

Academic Funding

- PI, National Natural Science Foundation of China #12301358 (国家自然科学基金-青年项目), 项目名称: Statistical modeling for complicated count time series with multivariate, non-stationarity and long-memory and its application in Covid-19 pandemic, 2024.01-2026.12.
- Participant. National Natural Science Foundation of China #42450275 (国家自然科学基金-原创探索项目, 参与人, 个人经费 50 万), 项目名称: 基于多模态数据融合计算的城市安全与舆情智能分析, 2025.01-2027.12.
- Participant. 北京市自然科学基金-青年科学基金滚动支持项目(个人经费 10 万), 项目名称: 融合混合专家与交叉注意力机制构建多模态胰腺囊性肿瘤治疗选择与疗效预测一体化模型, 2026.07-2029.06.

教学项目

- 武汉大学统计学一流本科专业建设项目, 子项目负责人—《时间序列分析》课程体系建设, 2024-2025, 2 万, 结项.

*Corresponding Author

Working paper

1. **Xu, X.*** & Taniguchi, M. “General Directional Distributions from Long-Memory Spectra and Applications”. 2026.
2. Jiang, H., Ji, Z. & **Xu, X.*** “Estimating conditional covariance matrices dependent on exogenous variables through a latent factor model”. 2026
3. **Xu, X.** & Zhu, F. K. “Heavy-tailed long-memory integer-valued time series modeling”. 2026

Preprint

1. Chen, M., Tang, S., Chen, L.*, et. al., **Xu, X.***, Kwan, M.* (2026+) “Birds of a feather”? The divergence in urban social mobility is widening. *Submitted*.
2. Yang J., Jiang, H.* & **Xu, X.** (2026+) “A nonparametric approach to the inference of the phase difference in time series”. *Submitted*.
3. Qing, H. & **Xu, X.*** (2026+) “Estimating mixed memberships in multi-layer networks”. *Submitted*.
4. Wang, Z., **Xu, X.** & Yue, M.* (2026+) “Sparse boosting for high dimensional AFT longitudinal joint modeling with right censored survival data”, *Submitted*.
5. Liu, P., Tee, C.W. & **Xu, X.*** (2026+) “Horizon-level stochastic optimization for multi-period portfolio decisions”. *Submitted*.
6. Liu, P. & **Xu, X.*** (2026+) “A functional vector autoregressive framework for dynamic portfolio optimization”. *Submitted*.
7. Qing, H. & **Xu, X.*** (2026+) “A signed network perspective of the stock market”. *Submitted*.
8. Dong, J., Ma, X.* & **Xu, X.** (2026+) “Profiled transfer learning for high dimensional quantile regression”. *Submitted*.
9. Chen, M., Chen, L.*, Lu, D., Han, M., **Xu, X.** and et al. (2026+) “Post-pandemic human mobility reveals fractal social stratification ”. *Submitted*.

Book

1. Goto, Y., Nagahata, H., Taniguchi, M., Monti A.C., & **Xu, X.** (2023) *ANOVA with Dependent Errors*. Springer Singapore. <https://doi.org/10.1007/978-981-99-4172-8>

Publications

1. Wang, Z., **Xu, X.**, Xie, X., & Yue, M*. (2026). Sparse boosting approach for high-dimensional Granger causality learning in multivariate time series. *Communications in Statistics - Simulation and Computation*, 1–23. <https://doi.org/10.1080/03610918.2026.2669363>
2. Qing, H. & **Xu, X.*** (2026) Finding core balanced modules in statistically validated stock networks. *Expert Systems with Applications*, 320, 132236. <https://doi.org/10.1016/j.eswa.2026.132236>
3. Qing, H. & **Xu, X.*** (2025) Latent class analysis with arbitrary-distribution responses. *Entropy*, 27(8), 866. <https://doi.org/10.3390/e27080866>
4. **Xu, X.***, Zhang, Y., Liu, Y., Goto, Y., Taniguchi, M., & Chen, Y. (2025) Long-memory log-linear zero-inflated generalized Poisson autoregression for Covid-19 pandemic modeling. *Statistica Sinica*. 35, 505-532. <https://doi.org/10.5705/ss.202022-0148w>
5. **Xu, X.***, Liu, Y. & Taniguchi, M. (2024) Second-order robustness for time series inference. *Statistical Inference for Stochastic Processes*. 27, 213–225 <https://doi.org/10.1007/s11203-023-09296-w>
6. **Xu, X.**, Chen, Y.*, Zhang, G. & Koch, T. (2024). Modelling functional time series and mixed-type predictors with partially functional autoregression. *Journal of Business and Economic Statistics*. 42(2), 349–366. <https://doi.org/10.1080/07350015.2021.2011299>
7. **Xu, X.***, Taniguchi, M., & Murata, N. (2023). UMVU Estimation for Time Series. In *Research Papers in Statistical Inference for Time Series and Related Models: Essays in Honor of Masanobu Taniguchi* (pp. 555-564). Singapore: Springer. https://doi.org/10.1007/978-981-99-0803-5_25

8. Goto, Y., Suzuki, K., **Xu, X.***, & Taniguchi, M. (2023) Tests for the existence of group effects and interactions for two-way models with dependent errors. *The Annals of the Institute of Statistical Mathematics*. 75, 511–532. <https://doi.org/10.1007/s10463-022-00853-3>
9. **Xu, X.***, Liu, Y., & Taniguchi, M. (2023) Higher-order asymptotics of minimax estimators for time series. *Journal of Time Series Analysis*. 44(2): 247-257. <https://doi.org/10.1111/jtsa.12661>
10. Petkovic, M*, Chen, Y., Gamrath, I., Gotzes, U., Hadjidimitrou, N. S., Zittel, J., **Xu, X.**, & Koch, T. (2022). A hybrid approach for high precision prediction of gas flows. *Energy Systems*. 13(2): 383-408. <https://doi.org/10.1007/s12667-021-00466-4>
11. **Xu, X.***, Li, Z., & Taniguchi, M. (2022) Comparison between the exact likelihood and Whittle likelihood for moving average processes. *STATISTICA*. 82(1) 3-13. <https://doi.org/10.6092/issn.1973-2201/13609>
12. **Xu, X.**, Kou, S.* & Chen, Y. (2021). Discussion on “Text Selection”. *Journal of Business and Economic Statistics*. 39(4), 883-887. <https://doi.org/10.1080/07350015.2021.1942890>
13. **Xu, X.***, & Zakiyeva, N. (2021). Nonlinear network autoregressive model with application to natural gas network forecasting. *Scientiae Mathematicae Japonicae*, e 2020 33 2020-7 (in Editione Electronica).
14. **Xu, X.**, Chen, Y., Chen, C. W. S.*, & Lin, X. (2020). Adaptive log-linear zero-inflated generalized Poisson autoregressive model with applications to crime counts. *The Annals of Applied Statistics*. 14(3): 1493-1515. <https://doi.org/10.1214/20-AOAS1360>
15. Chen, Y., Koch, T., Lim, K. G., **Xu, X.**, & Zakiyeva, N*.(alphabetical)(2020). A review study of functional autoregressive models with application to energy forecasting. *Wiley Interdisciplinary Reviews: Computational Statistics*, e1525. <https://doi.org/10.1002/wics.1525>

Thesis:

“Statistical modeling for high-dimensional and non-stationary time series”. Ph.D. thesis, NUS, Supervised by A/P Ying Chen, March 2020.

“Some features of the spread of epidemics on a random graph”. B.Sc. thesis, USTC, Supervised by Prof Zhi-Shui Hu, July 2015.

Visiting Experience

- Visiting to School of Business and Economics, Humboldt University of Berlin. Germany May - June 2016
 - Supervise a master student for a project of face recognition; Search dataset from website and do data processing; Use PCA and machine learning methods (SVM, LDA, etc.) to do classification.

Talks in Conference and Workshop

- 2025 年 8 月 17-20 号, 中国现场统计研究会旅游大数据分会、全国工业统计学教学研究会数字经济与区块链技术协会暨 2025 年学术年会, 中国, 合肥
 - Talk: *Multi-Period Portfolio Allocation: A One-Shot Stochastic Optimization Approach*
- 2025 年 7 月 31-8 月 2 号, 2025 年计量金融国际会议, 新加坡, 新加坡国立大学
 - Talk: *Multi-Period Portfolio Allocation: A One-Shot Stochastic Optimization Approach*
- 2025 年 7 月 4-6 号, 中国科学技术大学统计前沿会议, 中国, 合肥
 - Talk: *Multi-Period Portfolio Allocation: A One-Shot Stochastic Optimization Approach*
- 2025 年 6 月 6-8 号, 中国现场统计研究会多元分析应用专业 2025 年学术年会, 中国, 重庆
 - Talk: *Long-memory Log-linear Zero-inflated Generalized Poisson Autoregression for COVID-19 Pandemic Modelling*
- 2025 年 1 月 10 号, 中国长春, 吉林大学
 - Talk: *Long-memory Log-linear Zero-inflated Generalized Poisson Autoregression for COVID-19 Pandemic Modelling*
- 14-18 March, 2023, NUS WASEDA Workshop 2023, NUS, Singapore
 - Talk: *Log-linear Zero-inflated Generalized Poisson Autoregression for Covid-19 Pandemic Modelling*
- 8-9 July, 2022, EAC-ISBA 2022 Conference, Online meeting, Feng Chia University, Taiwan.

- Talk: *Minimax estimation for time series in view of higher-order asymptotics*
- 28-31 March, 2022, Mathematical Society of Japan Spring Meeting 2022, Online meeting, Japan.
 - Talk: *Comparison between the exact likelihood and Whittle likelihood for moving average processes*
- 10-12 March, 2022, Otsu Seminar “Recent Developments in Time Series and Related Topics” In honour of Professor Masanobu Taniguchi on the occasion of his retirement, Japan
 - Higher order asymptotics of minimax estimators for time series
- 7-9 March, 2022, Waseda International Symposium “Topological Data Science, Causality, Analysis of Variance & Time Series”, Tokyo, Japan
 - Talk: *Long-memory Log-linear Zero-inflated Generalized Poisson Autoregression for COVID-19 Pandemic Modelling*
- 14-17 September, 2021, Mathematical Society of Japan Autumn Meeting 2021, Online meeting, Japan.
 - Talk: *Minimax estimation for time series in view of higher-order asymptotics.*
- 5-9 September, 2021, Japanese Joint Statistical Meeting 2021, Online meeting, Japan.
 - Talk: *Minimax estimation for time series in view of higher-order asymptotics.*
- 3-4 June, 2021, Annual meeting of the Japanese Society of Computational Statistics, Online meeting, Japan.
 - Talk: *Zero-inflated Generalized Poisson Autoregression for US Congress Speech Phrase Counts.*
- 19-23 March, 2021, Waseda Cherry Blossom Workshop on Topological Data Science, Tokyo, Japan.
 - Talk: *Adaptive log-linear zero-inflated generalized Poisson autoregressive model with applications to crime counts.*
- 20-22 December, 2019, Invited session “Advanced Statistical Modeling for Complex Data” in the 11th ICSA International Conference, Hang Zhou, China.
 - Talk: *Adaptive log-linear zero-inflated generalized Poisson autoregressive model to crime counts*
- July 27-August 1, 2019, The Joint Statistical Meetings 2019, Denver, USA.
 - Session: Applications in Surveys & Social Science - Contributed Papers. Talk: *Adaptive log-linear zero-inflated generalized Poisson autoregressive model with applications to crime counts*
- 18-23 August, 2019, The 62th ISI World Statistics Congress 2019, Kuala Lumpur, Malaysia.
 - Invited session: Computational Statistics and Application. Talk: *Regularized partially functional autoregressive model with application to high-resolution natural gas forecasting in Germany*
- 18 – 20 May, 2018, The International Conference on Frontiers of Data Science, Hang Zhou, China,
 - Talk: *Regularized partially functional autoregressive model with application to high-resolution natural gas forecasting in Germany*
- 12 -13 May, 2018, The 3rd PKU-NUS Annual International Conference on Finance and Economics, Beijing, China,
 - Talk: *Regularized partially functional autoregressive model with application to high-resolution natural gas forecasting in Germany*

Teaching Experience

Lecturer

- Probability and Mathematical Statistics (概率论与数理统计 Undergraduate course, Wuhan Univ.)
- Time Series Analysis (时间序列分析 Undergraduate course, Wuhan Univ.)
- Statistical Software (统计软件 Undergraduate course, Wuhan Univ.)
- Mathematical Statistics (数理统计 Graduate course, Wuhan Univ.)

TA:

- Statistics Learning II (undergraduate compulsory course, Statistics department, NUS). Jan 2019 – May 2019
- Business Analytics-Data & Decision (undergraduate compulsory course, Business School, NUS). Aug 2016 – Dec 2018
- Statistics for Life Science (undergraduate compulsory course, Statistics department, NUS). Jan 2016 – May 2016
- Mathematical Statistics (undergraduate compulsory course, Statistics department, NUS). Aug 2015 – Dec 2015